# Mathematical Foundations of Political Methodology

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# Why Do We Need to Learn Calculus

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These slides preview and help justify the study of calculus for political scientists.

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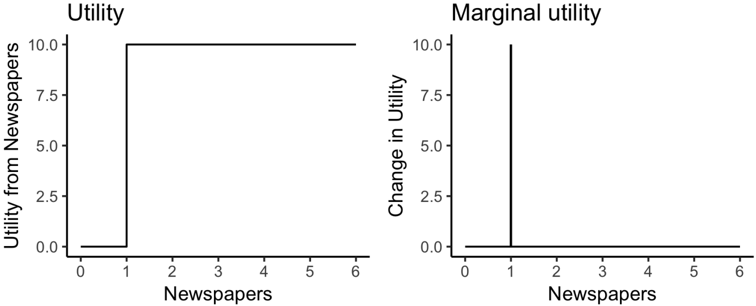
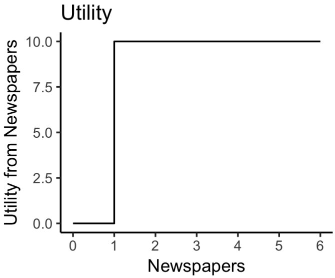
- The derivative refers to an instantaneous notion of change.
- The marginalist perspective reframes problems in terms of marginal costs and benefits: why do newspapers dispensers operate differently than candy dispensers?
- The first newspaper is valuable, the second and third less so. This is the notion of marginal utility.



# The benefits of being smooth



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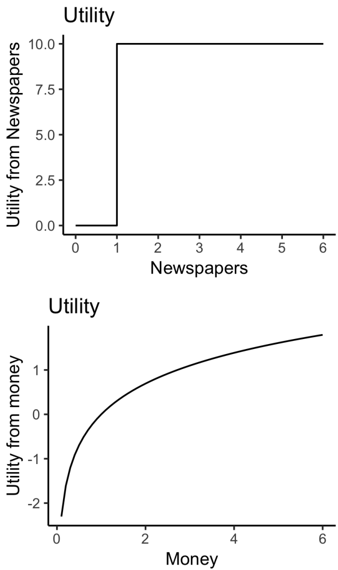

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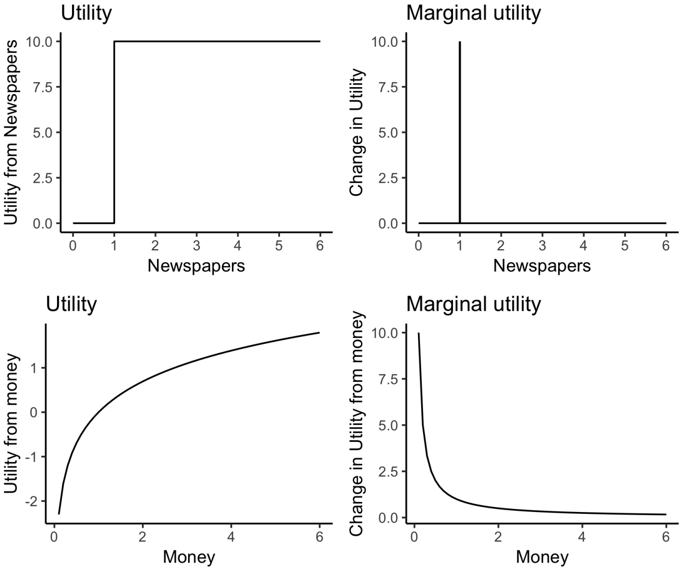

# Redistributive Distortionary Taxation

Suppose an individual's welfare (w) is 'quasi-linear':

$$w = c + V(x)$$

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- $x$ denotes hours devoted to leisure, where $V()$ is some (concave) function (like $\sqrt{x}$).

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There is only so much time in the day for work and play:

$$x + l \leq \alpha 16$$

where $\alpha$ is individual productivity.

# Solving Constrained Optimization Problems

$$\max_{x_1,\ x_2,\ldots} F(x_1,\ x_2,\ldots)$$

Such that: $x_1 \leq A, \quad x_2 \leq B \ldots.$

$$\mathcal{L}(x_1,\ x_2,\ldots\lambda_1,\lambda_2,\ldots) = F(x_1,\ x_2,\ldots) - \lambda_1(x_1 - A) - \lambda_2(x_2 - B)\ldots$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = 0$$

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$$x = \frac{1}{4}(y)^{-2} = f^{-1}(y)$$

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The difference between a 10 percent tax rate ($\tau = .1$) and a 20 percent tax rate ($\tau = .1$):

$$(\alpha 16 - \frac{1}{4}(1-.1)^{-2}) - (\alpha 16 - \frac{1}{4}(1-.2)^{-2})$$

$$\frac{1}{4}((.8)^{-2} - (.9)^{-2}) = .082$$

or .08 hours ($\approx$ 5 min) less work per day.

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The difference between a 50 percent tax rate and a 70 percent tax rate is 1.7 hours ($\approx$ 106min) less work per day.

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Now, suppose we have the SAT score in 2018, and suppose the distribution of SAT score is normal distribution with mean 1000, standard deviation 300.

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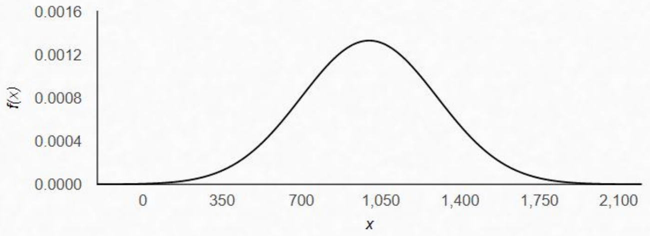
Now, let's see a picture of this normal distribution.

# Bell Curve




$\mu = E(X) = 1000 \qquad \sigma = SD(X) = 300 \qquad \sigma^2 = Var(X) = 90000$

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We need to compute this probability by:

$$Pr(x \geq 1500) = \int_{1500}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 0.04779$$

where $\sigma = 300$, $\mu = 1000$.